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|  <p><b>Innovative Academy</b><br/>Future Redefined</p> | <p><b>Syllabus –DevOps<br/>Engineer(BootCamp)</b></p> | <p>Website: <a href="https://innovativeacademy.in">https://innovativeacademy.in</a><br/>Email:<br/>Contact-Number:+918447712333</p> |
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### 01. Introduction to DevOps

- ✓ What is AWS
- ✓ What is Continuous Integration Video
- ✓ What is Continuous Delivery Video
- ✓ Tools Prerequisites Setup

### 02. Tools Installation Checklist

- ✓ Git CLI
- ✓ Git Bash for Windows
- ✓ Java – OpenJDK
- ✓ Maven
- ✓ Docker
- ✓ Docker-Compose
- ✓ AWS CLI
- ✓ Minikube, kubectl, Kops
- ✓ Terraform CLI
- ✓ Ansible

### 03. Introduction to Linux

- ✓ Introduction to Linux
- ✓ Commands and File systems
- ✓ More Commands ( mkdir, cp, mv, touch etc)
- ✓ Vim editor
- ✓ File Types
- ✓ Redirections
- ✓ Users and Groups
- ✓ File permissions
- ✓ Sudo
- ✓ Package Management
- ✓ Services
- ✓ Configuring Networking File Services (NFS & SAMBA)
- ✓ Configuring and Hosting Apache Webserver
- ✓ Process Management & Suitable Trouble-Shooting Concepts for Real-Time

### 04. Bash Scripting

- ✓ Introduction
- ✓ First script

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- ✓ Quotes
- ✓ User Input
- ✓ Command Substitution
- ✓ Command line arguments
- ✓ System Variables
- ✓ Exporting Variables
- ✓ Variables
- ✓ Variables with example
- ✓ Read with example
- ✓ Decision Making part1
- ✓ Nested Condition
- ✓ Decision Making part2
- ✓ Operators
- ✓ Decision Making with AND, OR
- ✓ Loops
- ✓ While Loops
- ✓ SSH & FOR LOOPS
- ✓ Web setup script with Decision
- ✓ Tomcat Setup
- ✓ Exercise FOR, IF etc

#### 05. Networking Fundamentals

- ✓ Understanding Networks & IP
- ✓ ISO
- ✓ Protocols, ports etc
- ✓ Networking Commands

#### 06. Aws solutions Architect For Devops

- ✓ Introduction
- ✓ EC2
- ✓ EBS
- ✓ EFS
- ✓ High-Availability with AWS Load Balancer
- ✓ Auto-scaling Group
- ✓ S3
- ✓ RDS

#### 07. AWS Cloud (Project1)– Setup Lift & Shift

- ✓ Introduction

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- ✓ Security Group & Key pairs
- ✓ EC2 Instances
- ✓ Build and Deploy Artifacts
- ✓ Load Balancer & DNS
- ✓ Auto scaling Group
- ✓ Validate & Summarize

#### 08. Re-Architecting WebApp on AWS Cloud With PAAS & SAAS

- ✓ Introduction
- ✓ RDS
- ✓ Elastic Cache
- ✓ Amazon MQ
- ✓ DB Initialization
- ✓ Beanstalk
- ✓ Build & Deploy Artifact
- ✓ Cloud front
- ✓ Validate and Summarize

#### 09. AWS CI&CD

- a. Introduction
- b. Beanstalk
- c. RDS & App Setup on Beanstalk
- d. Code Commit
- e. Code build
- f. Build, Deploy & Code Pipeline

#### 10. GIT for DevOps

- ✓ Introduction
- ✓ Versioning
- ✓ Branches & More
- ✓ Rollback
- ✓ SSH

#### 11. Build Tool for DevOps-Java-Maven

- ✓ Introduction
- ✓ Installing & setting with maven
- ✓ Running tests and building artifacts

#### 12. Continuous Integration Jenkins

- ✓ Introduction
- ✓ Installation

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- ✓ First Job
- ✓ Tools in Jenkins
- ✓ Plugins, Versioning & more
- ✓ Nexus Software Repo Setup
- ✓ Nexus Usage
- ✓ Notifications
- ✓ Code Analysis
- ✓ Build Pipeline & Triggers

### 13. Automation With Ansible

- ✓ Introduction
- ✓ Setup Ansible & Infra
- ✓ Inventory & Ping Module
- ✓ YAML & JSON
- ✓ Ad Hoc Commands
- ✓ Playbook & Modules
- ✓ Modules – Find, Use, Troubleshoot & Repeat
- ✓ Ansible Configuration
- ✓ Variables & Debug
- ✓ Group & Host Variables
- ✓ Fact Variables
- ✓ Decision Making, Loops & More
- ✓ Loops
- ✓ File, copy & template modules
- ✓ Roles
- ✓ Ansible for AWS

### 14. Continuous Delivery and Configuration Management [Jenkins plus Ansible]

- ✓ Introduction
- ✓ Continuous Integration
- ✓ Launch Staging Server
- ✓ Playbook for Tomcat Setup
- ✓ Playbook For Deployment
- ✓ Ansible on Jenkins
- ✓ Prod Deployment

### 15. Docker

- a. Introduction
- b. Docker Setup

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- c. Docker commands & concepts
- d. Docker volumes
- e. Building images
- f. Containerizing project

#### 16. Kubernetes

- a. 001 Introduction
- b. 002 Kubernetes Setup
- c. 003 Pods
- d. 004 Service
- e. 005 Replication Controller

#### 17. Java App Deployment on Kubernetes Cluster

- a. Introduction
- b. Spin K8s Cluster
- c. Source Code Overview
- d. Deploy Database On AWS RDS
- e. Tomcat Deployment, Service & Init containers
- f. Kubernetes Troubleshooting Ideas
- g. URL for Website & Wrap up

#### 18. Terraform Tutorials

- a. Introduction
- b. Variables – Exercise
- c. Provisioners – Exercise
- d. Output – Exercise
- e. Backend – Exercise
- f. Multi\_Resource – Exercise

#### 19. CloudFormation Tool Tutorial

- a. Introduction
- b. First Example
- c. Intrinsic Function
- d. Change Set
- e. More Intrinsic Functions
- f. Multiple Resources
- g. Mappings And Pseudo Parameters
- h. Parameters
- i. Outputs
- j. Init